

Control work, and the quarter review

Coordinates, vectors, the scalar product, symmetry and the vector line — one paper, then the map.

LESSON 17–18

2 × 40 MIN

GEOMETRY 11, NAZORAT ISHI 1

P3 · CHAPTER 9 REVIEW

LESSON CLOCK

3

42'

12'

20'

10'

3

Instructions	3 min
The paper	42 min
Answers	12 min
Rewrite	20 min
Concept map	10 min
Targets	3 min

LEARNING OBJECTIVES

- Apply the whole quarter in one assessment.
- Choose between the coordinate and the vector method.
- Classify each lost mark and rewrite the solution.
- Build a concept map of vectors in space.

TEXTBOOK

National	pp. 85–88
Cambridge	Pure Mathematics 2 & 3, pp. 209–212
Workbook	Control paper G1

01

Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS	FROM
1	Distance and midpoint of $A(1,-2,4)$ and $B(5,2,16)$	4	L1–3
2	Centre and radius of $x^2+y^2+z^2-4x+2y-4=0$	3	L1–3
3	$a = (2,-1,2)$, $b = (1,2,2)$: find $ a $, $a \cdot b$ and the angle	5	L7–8
4	Find t so that $(t,3,-1)$ and $(2,t,5)$ are perpendicular	3	L7–8
5	Are $A(1,0,2)$, $B(3,2,4)$, $C(6,5,7)$ collinear? Give $AB:BC$	4	L6
6	Vector equation of the line through $(2,-1,3)$ and $(4,3,7)$; is $(6,7,11)$ on it?	5	L14–16
7	Count the planes and axes of symmetry of a cube and of a $3 \times 3 \times 8$ cuboid	6	L10–11

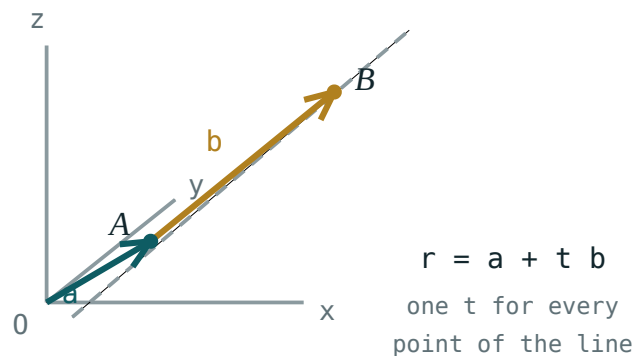
Q3 AND Q6 CARRY THE METHOD MARKS

Q3 gives 2 for the angle formula written correctly before any numbers go in. Q6 gives 2 for the direction vector and 3 for the test — a stated answer with no substitution scores one.

The concept map

Six boxes, links as sentences:

- **coordinates** → **distance** — “Pythagoras twice”
- **coordinates** → **position vectors** — “the same three numbers, read as a displacement”
- **vectors** → **collinear / coplanar** — “one scalar, or two”
- **vectors** → **scalar product** — “a number that carries the angle”
- **scalar product** → **perpendicularity** — “zero means a right angle”
- **point + direction** → **the line** — “ $r = a + td$ generates every point”



The picture that ties the last three boxes together.

Looking forward

Quarter II is polyhedra, prisms and cylinders — surface area and volume. Every angle in it is found by the scalar product of this quarter, and every distance by the formula of Lesson 1–3. Neither is re-taught.

ONE HABIT TO CARRY FORWARD

Write the vector before the arithmetic. $AB = b - a$, then compute. Learners who go straight to numbers lose the sign of the direction and, with it, the angle.

02 Worked examples

EXAMPLE 1 Model answer, Q3: $a = (2, -1, 2)$, $b = (1, 2, 2)$.

$$① \quad |a| = \sqrt{4+1+4} = 3$$

$$② \quad |b| = \sqrt{1+4+4} = 3$$

$$③ \quad a \cdot b = 2 - 2 + 4 = 4$$

$$④ \quad \cos \theta = \frac{4}{9} \Rightarrow \theta \approx 63.6^\circ$$

ANSWER

$$|a| = 3, a \cdot b = 4, \theta \approx 63.6^\circ$$

EXAMPLE 2 Model answer, Q6: line through $(2, -1, 3)$ and $(4, 3, 7)$; is $(6, 7, 11)$ on it?

$$① \quad d = (2, 4, 4), \text{ simplify to } (1, 2, 2).$$

$$② \quad r = (2, -1, 3) + t(1, 2, 2)$$

$$③ \quad 2 + t = 6 \Rightarrow t = 4$$

$$④ \quad \text{Check: } -1 + 8 = 7 \checkmark, 3 + 8 = 11 \checkmark$$

ANSWER

Yes, at $t = 4$

EXAMPLE 3 Model answer, Q2: centre and radius of $x^2+y^2+z^2-4x+2y-4=0$.

① $(x-2)^2 - 4$

② $(y+1)^2 - 1$

③ $(x-2)^2 + (y+1)^2 + z^2 = 9$

④ Read off.

ANSWER

centre $(2, -1, 0)$, radius 3

03 Interactive model

Work Q3 and Q6 on the board before the rewrite hour begins.

The quarter in ten questions

Distance $A(1, -2, 4)$ to $B(5, 2, 16)$:

Midpoint of those points:

Radius of $(x-2)^2 + (y+1)^2 + z^2 = 9$:

$(2, -1, 2) \cdot (1, 2, 2)$:

4

0

8

-4

The angle between them:

30°

45°

$\approx 63.6^\circ$

90°

$(t, 3, -1) \perp (2, t, 5)$ gives t :

1

-1

5

-5

$A(1, 0, 2)$, $B(3, 2, 4)$, $C(6, 5, 7)$ are:

collinear

not collinear

coplanar only

perpendicular

Direction of the line through $(2, -1, 3)$ and $(4, 3, 7)$:

$(1, 2, 2)$

$(2, 2, 4)$

$(1, 1, 1)$

$(2, -1, 3)$

Planes of symmetry of a $3 \times 3 \times 8$ cuboid:

3

4

5

9

Two non-parallel lines with an inconsistent system are:

parallel

intersecting

skew

identical

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Concept map	Tushunchalar xaritasi	Карта понятий
Justification	Asoslash	Обоснование
Method choice	Usul tanlash	Выбор метода
Self-assessment	O'z-o'zini baholash	Самооценка
Revision	Takrorlash	Повторение
Target	Maqsad	Цель
Command word	Topshiriq so'zi	Командное слово

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The most efficient test for a right angle in space:

measure it

the cosine rule

$a \cdot b = 0$

Pythagoras

Two lines are skew when:

directions parallel

the system is inconsistent and directions differ

they meet

they are equal

A vector equation of a line is unique:

always

never

only in 2-D

only through the origin

Quarter II begins with:

the derivative

polyhedra and volume

probability

trigonometry

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Distance $A(1, -2, 4)$ to $B(5, 2, 16)$

ANSWER 13

2. Midpoint of those

ANSWER $(3, 0, 10)$

3. Radius of $x^2 + y^2 + z^2 = 49$

ANSWER 7

4. $(2, -1, 2) \cdot (1, 2, 2)$

ANSWER 4

5. $|(2, -1, 2)|$

ANSWER 3

6. Planes of a cube

ANSWER 9

7. Direction through $(2, -1, 3)$ and $(4, 3, 7)$

ANSWER $(1, 2, 2)$

Medium · the standard question

7 PROBLEMS

1. Centre and radius of $x^2+y^2+z^2-4x+2y-4=0$

ANSWER $(2, -1, 0), 3$

2. Angle between $(2, -1, 2)$ and $(1, 2, 2)$

ANSWER $\approx 63.6^\circ$

3. Find t : $(t, 3, -1) \perp (2, t, 5)$

ANSWER $t = 1$

4. Are $A(1, 0, 2)$, $B(3, 2, 4)$, $C(6, 5, 7)$ collinear? $AB:BC$

ANSWER yes, $2:3$

5. Is $(6, 7, 11)$ on $r = (2, -1, 3) + t(1, 2, 2)$?

ANSWER yes, $t = 4$

6. Planes of a $3 \times 3 \times 8$ cuboid

ANSWER 5

7. Axes of a cube

ANSWER 13

Hard · stretch and reason

7 PROBLEMS

1. Find the point of $r = (2, -1, 3) + t(1, 2, 2)$ closest to the origin

ANSWER $(\frac{16}{9}, -\frac{13}{9}, \frac{19}{9})$

2. Show two given lines are skew and find the angle between them

ANSWER Inconsistent system; angle from $|d_1 \cdot d_2|$

3. Find the sphere through $(0, 0, 0)$, $(2, 0, 0)$, $(0, 2, 0)$, $(0, 0, 2)$

ANSWER $(x-1)^2 + (y-1)^2 + (z-1)^2 = 3$

4. Prove the diagonals of a cube meet at its centre

ANSWER All four have midpoint $(\frac{a}{2}, \frac{a}{2}, \frac{a}{2})$

5. Angle between two space diagonals of a cube

ANSWER $\approx 70.5^\circ$

6. Find λ so $(1, 2, \lambda)$ is coplanar with $(1, 0, 1)$ and $(0, 1, 1)$

ANSWER $\lambda = 3$

7. Where does $r = (1, 1, 1) + t(2, -1, 3)$ meet the plane $x = 5$?

ANSWER $(5, -1, 7)$

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter II.

1. Rewrite in full every control-work question that lost a mark.
2. Finish the concept map with all six links written as sentences.
3. Find the vector equation of the line through $(0, 1, -2)$ and $(4, 5, 6)$, and test whether $(6, 7, 10)$ is on it.

4. Write your target for Quarter II in one checkable sentence, and date it.